Weak-to-Strong Generalization Take-Home Interview Instructions

Extending Weak-to-Strong Generalization

OpenAI's Weak-to-Strong Generalization (W2SG) paper studies whether or not we should expect superhuman AIs to generalize appropriately from human-level supervision. They do this by comparing whether or not stronger AI students can learn to generalize from the supervision of weaker AI teachers. They find positive results in some domains, including generalization improving with scale, but this doesn't hold for all domains, most notably for the ChatGPT reward modeling task.

One important extension of this work could be to study how generalization and scaling trends are affected by including a small amount of strong supervision when training the student. This is plausibly more realistic – it's more likely that we'll have a supervision "budget", which we may choose to spend on a combination of "weak" and "strong" labels. The strong labels might be constructed with a scalable oversight protocol that amplifies the strength of an ordinary human supervisor, but it'll likely be the case that we can't afford to use this procedure to amplify all labels, since the oversight protocol might be quite expensive.

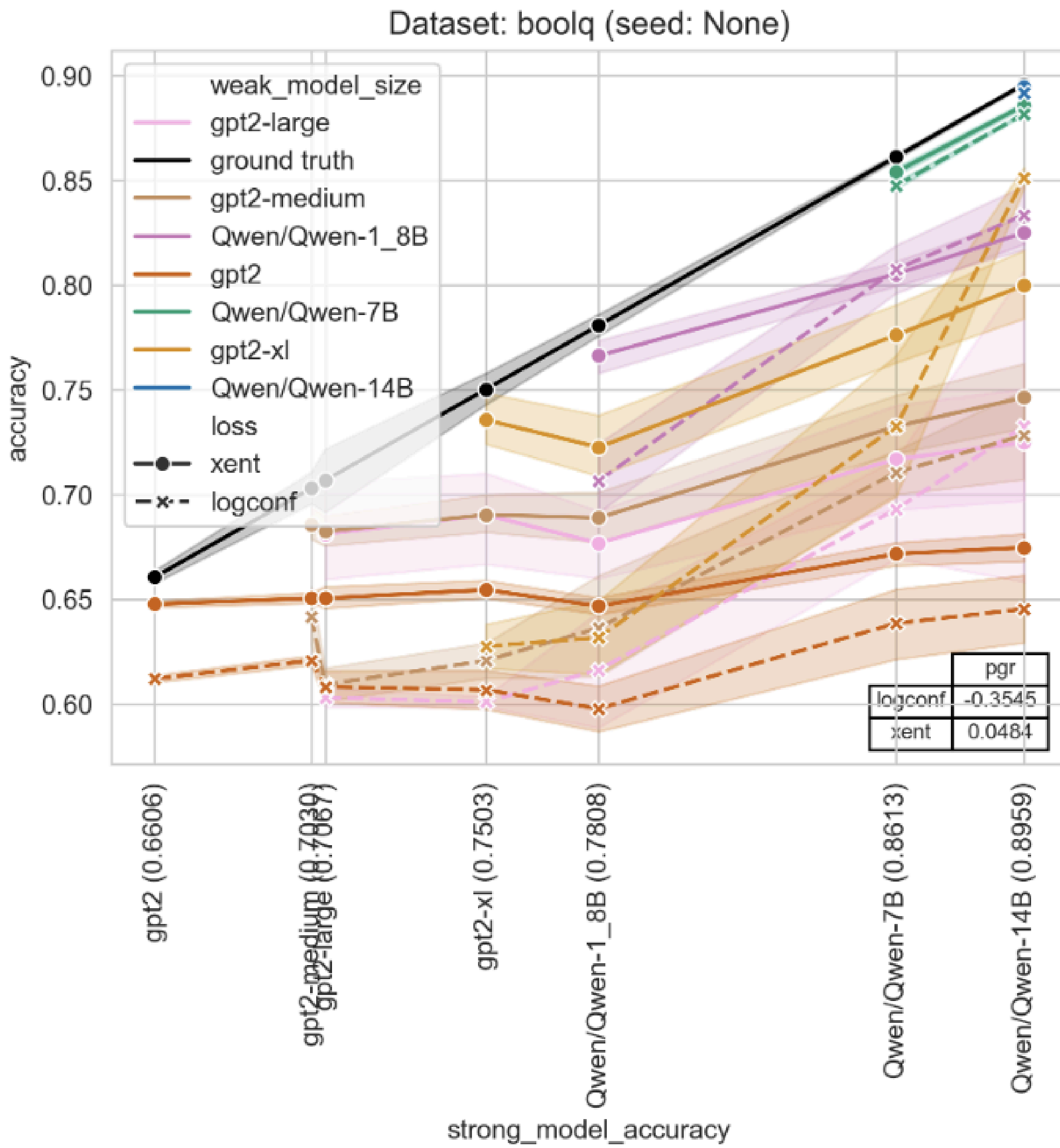
We'd be interested in seeing whether including a small fraction of ground-truth labels (as a stand-in for strong supervision) significantly helps W2SG for a given natural language task. OpenAI has an open-source repository which should allow you to conduct your own W2SG experiments on open-source/weight models. It should be reasonably straightforward to extend their implementation to support training of a student on some fraction of ground-truth labels (say, 25%) in addition to weak teacher predictions.

We leave most of the details of how to implement this, and most importantly, how to get this to work, to you! As a sanity check, it might make sense to start by including a quite large fraction of ground-truth labels and then to gradually decrease the fraction to see if your strategy can maintain high sample efficiency. You should feel free to mix in the labels however you wish during training and to modify the model architectures if you decide that doing so might be helpful.

At minimum, for any given method you try, we're expecting:

- A version of the below plot, except that it's restricted to the GPT-2 series of models
    - **Please make sure to replicate this exact plot format, including reporting the median PGR across the whole sweep of models, since it's helpful for standardizing results.**
- Trained on the BoolQ dataset included by default in OpenAI's open-source repo

- And trained with **25%** of ground-truth labels instead of purely trained on the predictions of the weak teacher.
- Extending the results to larger models (larger than GPT-2-xl) isn't required.
    - For the purpose of your experiments, no models other than the GPT2 family exist at all :)



Evaluation criteria

We're interested in seeing you:

1. Try out as many different approaches/ideas/experiments as possible (even if many of them fail, but are plausible/interesting/promising ideas) while aiming for good generalization and maintaining experimental rigor.
2. Make a clear presentation of your findings, **no more than 20 minutes long**, with a set of slides, at the end of the takehome period.

Deliverables
1. A set of presentation slides to accompany your live presentation (should take no more than 20 minutes to present).
2. A time log (more detail on this below).
3. A link to a private GitHub repository where you've forked the original code and extended it for your own experiments.

Please share all of these with your interviewers before the presentation at the end of the 72 hour period.

Notes:
1. Please log how many hours you spend on the task, so that we can better understand how much time you spent on the task and better evaluate your results.
2. You are allowed to use whatever productivity tools you'd like (e.g. code autocomplete, LLM/chatbot consultation, etc.) to assist you on the project.
    a. However, **please do not talk with other people about the interview during or after the interview**!
3. For compute, feel free to use any compute provider that you're comfortable with and spend up to $2000 – we'll be able to fully reimburse you for this amount.
    a. Previous candidates have used providers like LambdaLabs or RunPod.
    b. Instructions for getting set up with LambdaLabs: https://lambdalabs.com/blog/getting-started-with-lambda-cloud-gpu-instances
        i. We generally recommend trying to grab an 8xH100 if possible, but an 8xA100 or 4xH100 should also be totally sufficient.